

## 11.5 Power and noncentrality

A test is judged by two probabilities. Its size, the probability of rejecting a true hypothesis, is fixed by the critical value. Its **power**, the probability of rejecting a false one, depends on how false the hypothesis is. For the  $F$  test the whole dependence runs through the single number  $\gamma$ , and that makes power calculations unusually transparent. This section states the power function, interprets the noncentrality, uses it to plan the size of a study, and asks how sensitive the region test of [Section 11.1](#) actually was. It ends with a popular calculation that should not be made.

### 11.5.1 The power function

#### Theorem 11.5.1 — Power of the $F$ test

Assume Eq. 7.3.1 with  $r < n$ , and let  $H : \Lambda^\top \beta = \mathbf{d}$  be testable with  $q = \text{rank}(\Lambda)$  (or let  $H$  be a reduced model with  $q = r - r_0$ ). The probability that the level- $\alpha$   $F$  test rejects is

$$\pi(\gamma) = \Pr\{F(q, n - r, \gamma) > F_\alpha(q, n - r)\}, \quad (11.5.1)$$

where  $\gamma$  is the noncentrality of Eq. 11.3.2, or of Eq. 11.1.2 for a reduced model. Moreover:

- (a)  $\pi$  depends on  $(\beta, \sigma^2)$  only through  $\gamma$ . It is continuous and strictly increasing, with  $\pi(0) = \alpha$  and  $\pi(\gamma) \rightarrow 1$  as  $\gamma \rightarrow \infty$ . In particular the test is *unbiased*: its power exceeds  $\alpha$  at every alternative.
- (b)  $\sigma^2\gamma$  is the squared distance from the mean vector  $\mathbf{X}\beta$  to the set of mean vectors allowed by  $H$ :

$$\sigma^2\gamma = \min_{\Lambda^\top \mathbf{b} = \mathbf{d}} \|\mathbf{X}\beta - \mathbf{X}\mathbf{b}\|^2 = \min_{\Lambda^\top \mathbf{b} = \mathbf{d}} \sum_{i=1}^n (\mathbf{x}_{(i)}^\top \beta - \mathbf{x}_{(i)}^\top \mathbf{b})^2.$$

- (c) If further observations are added, with the same  $\beta, \sigma^2$  and hypothesis, the noncentrality does not decrease. If the whole design is replicated  $k$  times,  $\gamma$  is multiplied by  $k$ .

*Proof.* The distribution of  $F_H$  is Theorem 11.3.2(c), which gives [Eq. 11.5.1](#). (a) is Theorem 4.1.4 with  $c = F_\alpha(q, n - r)$ . For (b), Theorem 11.3.2(b) applied to the data vector  $\mathbf{X}\beta$ , for which  $\beta$  is a least squares estimate and the residual sum of squares is zero, shows that  $\min_{\Lambda^\top \mathbf{b} = \mathbf{d}} \|\mathbf{X}\beta - \mathbf{X}\mathbf{b}\|^2 = (\Lambda^\top \beta - \mathbf{d})^\top \mathbf{W}^- (\Lambda^\top \beta - \mathbf{d}) = \sigma^2\gamma$ . For (c), adding rows to  $\mathbf{X}$  enlarges  $\mathcal{C}(\mathbf{X}^\top)$ , so  $H$  remains testable (the constraint set  $\{\mathbf{b} : \Lambda^\top \mathbf{b} = \mathbf{d}\}$  is unchanged), and the sum in (b) acquires extra nonnegative terms for every  $\mathbf{b}$ . The minimum of a larger function over the same set is at least as large. Replicating the design  $k$  times multiplies every sum of squares in (b) by  $k$ , and hence the minimum. ■

Part (b) is the reading of  $\gamma$  to remember. The noncentrality is the smallest total squared

discrepancy, over all observations, between the true mean and a mean the hypothesis allows, in units of the error variance. It is large when the hypothesis is badly wrong, when  $\sigma^2$  is small, and when there are many observations at which the discrepancy shows. Part (c) makes the last point precise. Note that adding observations also increases  $n - r$ , which helps as well, and that the sum in (b) does *not* depend on the parameterization.

Two consequences follow at once. First, for a fixed alternative,  $\gamma$  grows in proportion to the number of replicates, so the power tends to one: *every fixed departure from the hypothesis is detected with certainty in large enough samples*. A significant result from a very large sample may therefore reflect a departure too small to matter, and effect sizes, not p-values, answer the question “how much?”. Second, the ratio  $\gamma/n$  is a natural *effect size*, the average squared discrepancy per observation relative to  $\sigma^2$ . It is closely related to the effect size  $f^2$  of Cohen’s power tables, which are widely used in the behavioural sciences.

## 11.5.2 Planning a study

Before the data are collected, Eq. 11.5.1 turns the question “how many observations?” into arithmetic. One specifies the smallest departure worth detecting, a guess at  $\sigma$ , the level and the desired power, and solves for  $n$ . The difficulty is the first ingredient. A departure is a whole vector  $\mathbf{A}^\top \boldsymbol{\beta} - \mathbf{d}$ , and different vectors of the same apparent size give different  $\gamma$ . For the comparison of group means there is a clean way out.

### Proposition 11.5.2 — Least and most favourable configurations

In the balanced one-way layout with  $g \geq 2$  groups of  $m$  observations, group means  $\mu_1, \dots, \mu_g$  and common variance  $\sigma^2$ , the noncentrality of the  $F$  test of equal means is

$$\gamma = \frac{m}{\sigma^2} \sum_{i=1}^g (\mu_i - \bar{\mu})^2, \quad \bar{\mu} = \frac{1}{g} \sum_i \mu_i.$$

Among configurations whose range  $\max_i \mu_i - \min_i \mu_i$  equals  $\Delta$ ,

$$\frac{m\Delta^2}{2\sigma^2} \leq \gamma \leq \frac{mg\Delta^2}{4\sigma^2}.$$

The lower bound is attained exactly when one mean is at each extreme and all others are at the midpoint. For even  $g$ , the upper bound is attained exactly when half the means are at each extreme.

*Proof.* The full model has one indicator per group. The reduced model is  $\mathcal{C}(\mathbf{1})$ , so by Theorem 11.1.1(c),  $\sigma^2\gamma = \|(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\boldsymbol{\mu}\|^2$ , where  $\boldsymbol{\mu}$  repeats  $\mu_i$  on the  $m$  observations of group  $i$ . This is  $m \sum_i (\mu_i - \bar{\mu})^2$ . Let  $\mu_{\max}$  and  $\mu_{\min}$  be two groups at the extremes and put  $a = \mu_{\max} - \bar{\mu} \geq 0$ ,  $b = \bar{\mu} - \mu_{\min} \geq 0$ , so  $a + b = \Delta$ . Dropping the other groups from the sum,

$$\sum_i (\mu_i - \bar{\mu})^2 \geq a^2 + b^2 \geq \frac{1}{2}(a + b)^2 = \frac{1}{2}\Delta^2,$$

with equality in the first step iff every other mean equals  $\bar{\mu}$ , and in the second iff  $a = b$ . If all other means equal  $\bar{\mu}$ , then  $\bar{\mu}$  is also the average of the two extremes, so  $a = b$  holds automatically, and the equality case is the stated configuration. For the upper bound, let  $c$  be the midpoint of the range. Since  $\bar{\mu}$  minimizes  $t \mapsto \sum_i (\mu_i - t)^2$ ,  $\sum_i (\mu_i - \bar{\mu})^2 \leq$

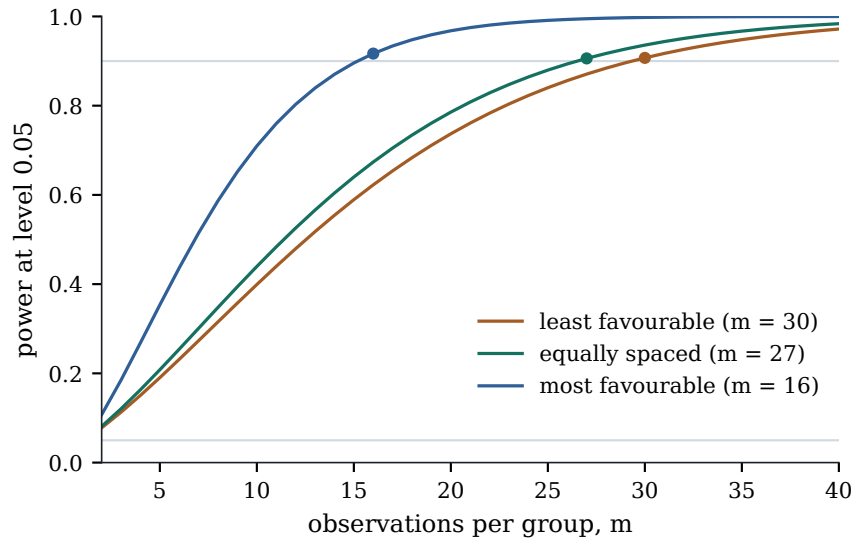


Figure 11.5.1: Power of the level-0.05  $F$  test of equal means for four groups of  $m$  observations, when the group means span one error standard deviation. The three curves are the least favourable configuration of Proposition 11.5.2, equally spaced means, and the most favourable configuration. Dots mark the smallest  $m$  giving power 0.9.

$\sum_i (\mu_i - c)^2 \leq g\Delta^2/4$ , because every  $|\mu_i - c| \leq \Delta/2$ . Equality requires every mean at an extreme and  $\bar{\mu} = c$ , which for even  $g$  means half at each end. ■

A design that has power 0.9 at the least favourable configuration has at least that power at every configuration with range  $\Delta$  or more. That is the guarantee a planner usually wants.

#### Example. (How many observations per group?)

Four treatments are to be compared at level 0.05, and a spread of one standard deviation between the best and the worst treatment ( $\Delta/\sigma = 1$ ) should be detected with probability at least 0.9. The noncentrality per observation per group is 0.500 for the least favourable configuration, 0.556 for equally spaced means, and 1.000 for the most favourable one (two means at each end). Searching over  $m$  with Eq. 11.5.1 gives

| Configuration    | $\gamma/m$ | smallest $m$ with<br>power $\geq 0.9$ | power at that $m$ |
|------------------|------------|---------------------------------------|-------------------|
| least favourable | 0.500      | 30                                    | 0.907             |
| equally spaced   | 0.556      | 27                                    | 0.906             |
| most favourable  | 1.000      | 16                                    | 0.917             |

To be safe against every configuration with range at least  $\sigma$ , the study needs 30 observations per group. A simulation of 20000 data sets at the least favourable configuration with  $m = 30$  rejects in a proportion 0.909 of them, against the computed 0.907. Figure 11.5.1 shows the whole power curves.

```
import numpy as np
from scipy import stats
```

```

def power(gamma, q, nu, alpha=0.05):
    """P{F(q, nu, gamma) > F_alpha(q, nu)}. """
    crit = stats.f.ppf(1 - alpha, q, nu)
    return stats.ncf.sf(crit, q, nu, gamma) if gamma > 0 else alpha

g, delta_over_sigma = 4, 1.0
patterns = {
    "least favourable": np.array([0, 0.5, 0.5, 1]),
    "equally spaced": np.array([0, 1 / 3, 2 / 3, 1]),
    "most favourable": np.array([0, 0, 1, 1]),
}

def gamma_oneway(m, means):
    """Noncentrality m * sum (mu_i - mu_bar)^2 / sigma^2 for m observations per
    group. """
    dev = (means - means.mean()) * delta_over_sigma
    return m * np.sum(dev ** 2)

for name, means in patterns.items():
    m = 2
    while power(gamma_oneway(m, means), g - 1, g * (m - 1)) < 0.9:
        m += 1
    print(f"{name:17s} m = {m} per group, power {power(gamma_oneway(m, means), g -
    1, g * (m - 1)):.3f}")

```

[Run this code online](#)

The same logic applies to any linear hypothesis. For a regression coefficient  $\beta_j$ , with the other regressors in the model,  $\sigma^2\gamma = \beta_j^2 \|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2$ , where  $\mathbf{M}_{(j)}$  projects onto the other columns. The design enters through the length of the part of  $\mathbf{x}_j$  that the other regressors cannot explain (Theorem 6.6.2). Spreading the values of  $x_j$  increases power, and collinearity with the other regressors destroys it. Adding a covariate that is unrelated to  $\mathbf{x}_j$  but explains part of the response reduces  $\sigma^2$ , and so increases  $\gamma$ , with little loss in the length of the residualized  $\mathbf{x}_j$ .

### 11.5.3 How sensitive was the region test?

Power calculations are also useful *after* a nonsignificant result, provided the alternatives are fixed by the scientific question and not by the data. For the region test of Example (Do regions matter?), consider the alternative in which the Northeast differs from the other three regions by  $\delta$  murders per 100,000, with everything else as in the reduced model. The mean vector is then  $\mathbf{X}_0\mathbf{c} + \delta\mathbf{d}_N$ , where  $\mathbf{d}_N$  is the Northeast indicator, and by Theorem 11.1.1(c),

$$\gamma = \frac{\delta^2 \|(\mathbf{I} - \mathbf{M}_0)\mathbf{d}_N\|^2}{\sigma^2}.$$

There are 9 Northeast states, but  $\|(\mathbf{I} - \mathbf{M}_0)\mathbf{d}_N\|^2 = 5.76$  only: the covariates already account for part of the Northeast's distinctiveness, and only the rest counts as evidence about regions. Setting  $\sigma$  at  $s = 1.423$  as a planning value, the power of the 5% test is

| $\delta$ | 0.5   | 1.0   | 1.5   | 2.0   |
|----------|-------|-------|-------|-------|
| power    | 0.091 | 0.241 | 0.506 | 0.778 |

and power 0.8 needs  $\delta = 2.05$ . A Northeast shortfall of 1.5 murders per 100,000, about the size estimated in Example (Regional hypotheses with an over-parameterized model) (1.379 below the other three regions), would have been missed about half the time. The nonsignificant verdict of Example (Do regions matter?) therefore says little about whether regional differences of that size exist.

```
import statsmodels.api as sm

data = sm.datasets.statcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWNNSSSWMMMMSSNSNMMSMWNNWNNSMMSWNNSSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
X0 = np.column_stack([np.ones(n), data["poverty"], data["single"], data["urban"]])
D = np.column_stack([(region == c).astype(float) for c in "NSW"])
X = np.column_stack([X0, D])

b0, *_ = np.linalg.lstsq(X0, D[:, 0], rcond=None)
k2 = np.sum((D[:, 0] - X0 @ b0) ** 2) # ||(I - M0) d_N||^2: gamma = delta^2 k2 / sigma^2
bf, *_ = np.linalg.lstsq(X, y, rcond=None)
sigma = np.sqrt(np.sum((y - X @ bf) ** 2) / (n - 7)) # planning value for sigma
for delta in (0.5, 1.0, 1.5, 2.0):
    print(f"Northeast lower by {delta}: gamma = {delta**2 * k2 / sigma**2:.2f}, "
          f"power = {power(delta**2 * k2 / sigma**2, 3, n - 7):.3f}")
```

*Run this code online*

The calculation uses the estimate  $s$  in place of  $\sigma$ , which is itself uncertain. A more careful version would repeat it for a range of plausible values of  $\sigma$ , or replace the whole exercise by a confidence region for the regional effects, which shows directly which differences the data rule out (Chapter 12).

## 11.5.4 Observed power

A common calculation takes the observed data, estimates the noncentrality from them, and reports the power “the test had” at that estimate. It is meant to show whether a nonsignificant result was due to low power. It cannot show that.

### Proposition 11.5.3 — Observed power is a function of the p-value

Fix  $q$ ,  $n - r$  and  $\alpha$ , and define the **observed power** as  $\pi(\hat{\gamma})$ , with  $\pi$  as in Eq. 11.5.1 and  $\hat{\gamma} = qF_{\text{obs}}$ . Then the observed power is a strictly decreasing function of the p-value of the  $F$  test. The same holds for any estimate  $\hat{\gamma} = h(F_{\text{obs}})$  with  $h$  strictly increasing and positive.

*Proof.* The p-value is  $1 - \Phi_F(F_{\text{obs}})$ , with  $\Phi_F$  the continuous, strictly increasing distribution function of  $F(q, n - r)$ , so  $F_{\text{obs}}$  is a strictly decreasing function of the p-value. The map  $F_{\text{obs}} \mapsto h(F_{\text{obs}})$  is strictly increasing, and  $\pi$  is strictly increasing by Theo-

rem 11.5.1(a). The composition is strictly decreasing in the p-value.

■

So observed power adds nothing to the p-value. In the setting of the region test ( $q = 3$ ,  $n - r = 43$ ), a p-value of 0.2 always corresponds to an observed power of 0.394, a p-value of exactly 0.05 to 0.637, and 0.01 to 0.830. “The test was not significant and its observed power was low” says the same thing twice. The informative calculations are the ones above: power at alternatives chosen for their scientific meaning, or a confidence region.

*Warning* (Reading a nonsignificant  $F$ ). A nonsignificant  $F$  statistic is weak evidence for the reduced model unless the test had high power against every alternative that matters. Whether it had is a question about the design and about which alternatives matter, answered by Eq. 11.5.1 at alternatives fixed in advance. It is not answered by the observed power, which merely restates the p-value, and it is not answered by the p-value being large. A p-value of 0.9 is no stronger evidence for  $H_0$  than one of 0.3 when the test has low power.

## 11.5.5 Exercises

### 11.5.5.1 A. Check your understanding

#### Exercise A1.

For the test of  $\beta_1 = 0$  in simple regression, show that  $\gamma = \beta_1^2 S_{xx} / \sigma^2$ , with  $S_{xx} = \sum_i (x_i - \bar{x})^2$ . With  $n = 20$  design points in  $[0, 1]$ , compare  $S_{xx}$  for equal spacing and for ten points at each end. What does the second design give up?

#### Exercise A2.

For the one-way layout with group sizes  $n_1, \dots, n_g$ , show that the noncentrality of the test of equal means is  $\gamma = \sum_i n_i (\mu_i - \bar{\mu}_w)^2 / \sigma^2$ , with  $\bar{\mu}_w = \sum_i n_i \mu_i / n$ .

### 11.5.5.2 B. Practice

#### Exercise B1.

Extend Proposition 11.5.2 to unequal group sizes: show that among configurations with range  $\Delta$ ,

$$\gamma \geq \frac{\Delta^2}{\sigma^2} \frac{1}{1/n_{(1)} + 1/n_{(2)}},$$

where  $n_{(1)} \leq n_{(2)}$  are the two smallest group sizes, and describe the configuration attaining it.

*Solution.* Let groups  $a$  and  $b$  carry the largest and smallest means. Dropping the other terms of Exercise (A2),  $\sigma^2 \gamma \geq n_a (\mu_a - \bar{\mu}_w)^2 + n_b (\mu_b - \bar{\mu}_w)^2 \geq \min_t \{n_a (\mu_a - t)^2 + n_b (\mu_b - t)^2\}$ . The minimum is at  $t = (n_a \mu_a + n_b \mu_b) / (n_a + n_b)$  and equals  $\Delta^2 n_a n_b / (n_a + n_b) = \Delta^2 / (1/n_a + 1/n_b)$ . This is smallest when  $a$  and  $b$  are the two smallest groups.

It is attained when those two groups carry the extremes and all other means equal  $(n_a\mu_a + n_b\mu_b)/(n_a + n_b)$ , which is then also  $\bar{\mu}_w$ . A departure is hardest to detect when it is confined to the groups with the fewest observations.

### Exercise B2.

Take the straight-line model with design points 0, 1, 2, 3, one observation each, and the hypothesis  $\beta_1 = 0$ . Compute  $\gamma$  as a multiple of  $\beta_1^2/\sigma^2$  and the power at  $\beta_1/\sigma = 1$ , level 0.05. Then compare (i) replicating the design twice, (ii) adding four observations at  $x = 3$ , and (iii) adding four observations at  $x = 1.5$ . Verify [Theorem 11.5.1\(c\)](#) in each case and explain the ranking.

### Exercise B3.

For  $q = 1$  and large  $n - r$ , the  $F$  test is approximately the test that rejects when  $|Z| > z_{\alpha/2}$ , with  $Z \sim N(\delta, 1)$ . Show that if the observed  $|z|$  equals  $z_{\alpha/2}$  exactly, the observed power  $\Pr\{|Z| > z_{\alpha/2}\}$  at  $\delta = z_{\alpha/2}$  is just above  $1/2$ . Conclude that a result on the edge of significance always has observed power about one half, whatever the design.

*Solution.* At  $\delta = z_{\alpha/2}$ ,  $\Pr\{|Z| > z_{\alpha/2}\} = \Pr\{Z > z_{\alpha/2}\} + \Pr\{Z < -z_{\alpha/2}\} = \frac{1}{2} + \Pr\{Z - \delta < -2z_{\alpha/2}\}$ , and the last probability is  $\Phi(-2z_{\alpha/2})$ , which is  $\Phi(-3.92) \approx 4 \times 10^{-5}$  for  $\alpha = 0.05$ . The value depends only on  $\alpha$ , not on  $n$ ,  $\sigma$  or the design, as [Proposition 11.5.3](#) predicts.

### 11.5.5.3 C. Going deeper

#### Exercise C1.

In the sensitivity calculation, let  $n_N$  be the number of Northeast states and  $R_N^2$  the coefficient of determination from regressing the Northeast indicator on the reduced model's columns (which include  $\mathbf{1}$ ). Show that  $\|(\mathbf{I} - \mathbf{M}_0)\mathbf{d}_N\|^2 = n_N(1 - n_N/n)(1 - R_N^2)$ . Interpret each factor, and relate the formula to Eq. 6.6.2.